

GE261004 DESIGN THINKING

Unit I – Introduction to Design Thinking · Presentation 2 of 6

Engineering Design vs Traditional Problem Solving

Well-structured and ill-structured problems · linear and iterative work · divergent and convergent thinking · the Double Diamond · solution-focused reasoning

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In this presentation

- The problems your other courses set
- Linear and iterative work
- The two approaches side by side
- Divergent and convergent thinking
- The Double Diamond
- Why engineers find divergence hard
- Analysis, synthesis and evaluation
- Problem-focused and solution-focused

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Two ways of solving

What changes when the problem is no longer handed to you

The problems your other courses set

- Every quantity you need is given – and nothing you do not need is
- There is one correct answer, and a marking scheme that already knows it
- The method is taught before the problem is set
- Your task is to execute that method accurately and completely

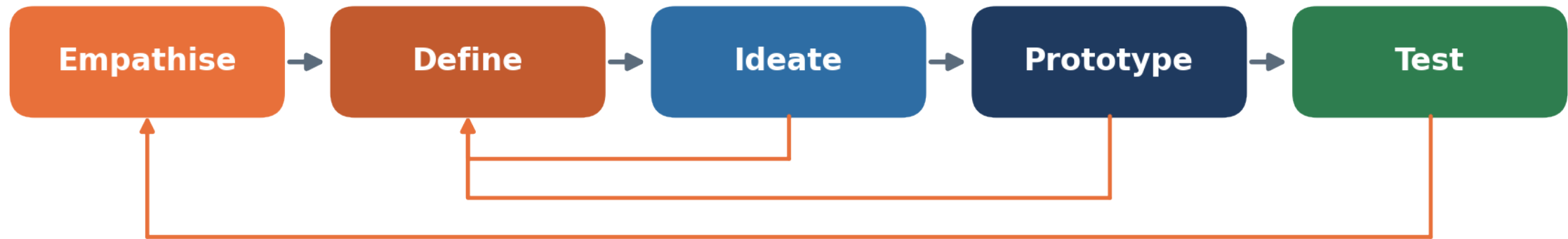
KEY POINT This is a necessary skill and this course does not replace it – but it is the second half of engineering, not the whole of it.

Linear and iterative work

Traditional linear problem solving



Design thinking – iterative



loop back whenever what you learn changes the problem

Traditional solving runs forward once; design thinking runs forward and back.

The two approaches side by side

	Traditional problem solving	Design thinking
The problem	Given, fully specified	Discovered, then reframed
Information	Supplied with the question	Gathered from users in the field
The answer	One correct solution	Many workable options, one chosen
Failure	To be avoided at all costs	To be caused early, cheaply and on purpose
It ends when	The answer is verified	The user says it works

Engineering practice needs both columns. Most graduates arrive strong on the left and untrained on the right.

2

Divergent and convergent thinking

Opening a problem up before closing it down – twice

Two thinking modes

Divergent thinking

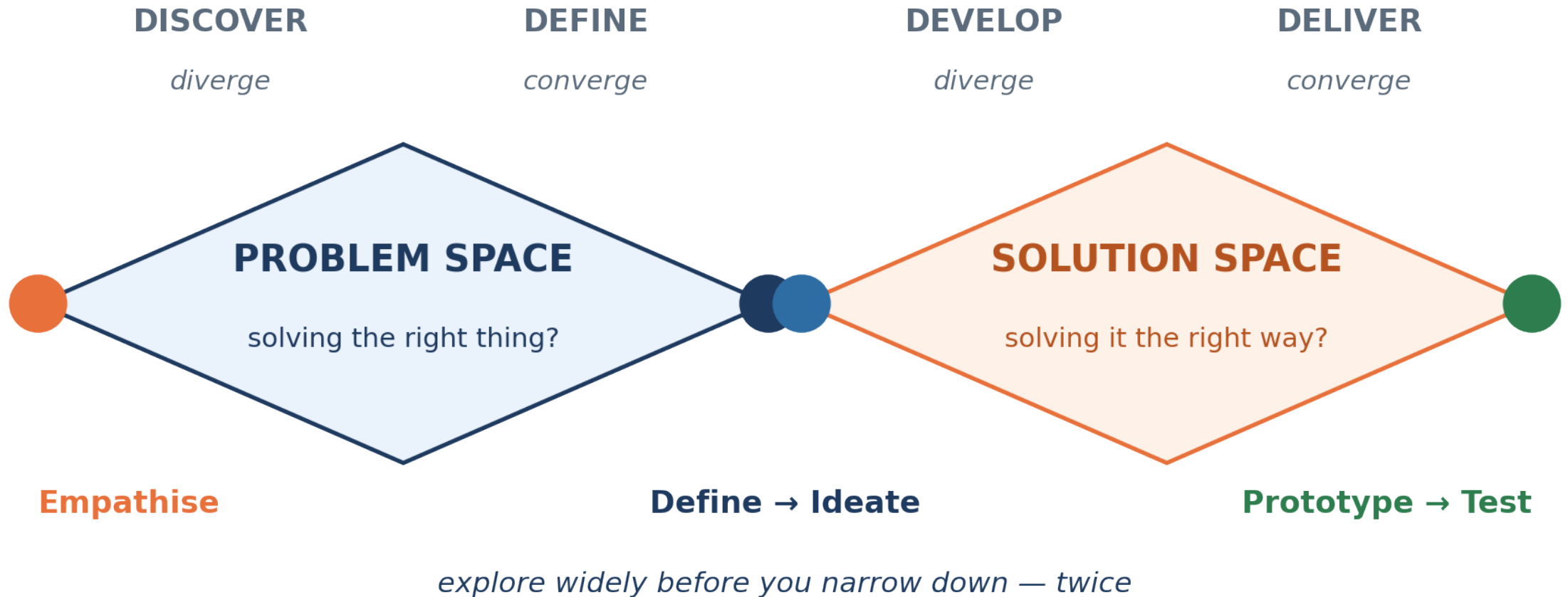
- Generate as many options as possible
- Defer judgement – no criticism yet
- Quantity first; quality is selected later
- Ends with a wide field of possibilities

Convergent thinking

- Compare, evaluate and eliminate
- Apply criteria, constraints and evidence
- Quality of judgement matters most
- Ends with one option to carry forward

KEY POINT Running both modes at once is what kills good ideas. Separate them in time, and say aloud which mode the team is in.

The Double Diamond



First find the right problem, then find the right solution.

Why engineers find divergence difficult

- Years of training reward the fastest route to the one correct answer
- Feasibility is judged instinctively, so weak-looking ideas are killed before anyone has had a chance to improve them
- **Premature convergence** – the first workable idea quietly becomes the only idea on the table
- Fear of looking foolish in front of peers suppresses exactly the unusual suggestions that are worth having

KEY POINT In a divergent session, “that will not work” is out of order. Write it down and judge it in the convergent half.

Three kinds of reasoning in design

Analysis

Break the problem into parts and study each one. What your other courses train you to do.

Synthesis

Put parts together into a whole that did not exist before. The designer's own move.

Evaluation

Judge the whole against criteria and evidence, then go back and change it.

KEY POINT Design alternates between all three. Analysis alone has never produced a new object.

Problem-focused and solution-focused

- **Scientists** study the problem thoroughly, then derive the answer
- **Designers** propose a rough solution early, then use it to understand the problem
- Nigel Cross found this pattern in almost every expert designer he studied
- A rough solution is a question put to the world: its failure tells you what the problem really was

KEY POINT A rough solution is a research instrument, not a commitment.

In class: the T-chart

- Take one familiar object – the campus water cooler, a bus stop, a lab manual
- Left column: write it as a traditional engineering brief, fully specified
- Right column: write the same situation as an ill-defined human problem
- Compare across teams – how many different problems did one object produce?

ACTIVITY Fifteen minutes in teams of four. One sheet per team; two teams present at the end of the hour.

THINK ABOUT IT

You have one week and no budget. How would you find out whether a problem is worth solving at all?

- Who would you talk to first, and what would you ask them?
- What could you observe without asking anyone anything?
- What evidence would convince you to walk away from the problem?
- What would you have to build in order to learn something useful?

Summary

- Classroom problems are given, closed and singly answered
- Real problems must first be discovered and reframed
- Design thinking loops until the problem stops changing
- Divergence opens a space, convergence closes it
- The Double Diamond applies that pair twice
- Designers reason from rough solutions, not only from problems

Next: Innovation and creativity

- What creativity means, and how it is measured
- The barriers that stop it, and creative confidence
- Invention and innovation – not the same thing
- Incremental, architectural, radical and disruptive innovation